

# **Software Requirements Specification**

for

## **Career & Lifelong Learning Pathway Application**

Cal Poly Humboldt – Software Engineering – CS 458

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# 1. Introduction

## 1.1 Purpose and Intended Audience

The purpose of this document is to detail the projected software requirements for the proposed *Career & Lifelong Learning Pathway (CaLLP)* application as clarified by the project's sponsor, Cyril Oberlander, Cal Poly Humboldt Library Dean. This document is intended for developers, program managers, and other interested parties who wish to ascertain the technical scope and driving philosophy behind this application.

## 1.2 Project Scope

The *CaLLP* application will serve as a centralized skill development platform in which both students and alumni can be dynamically connected to learning resources specifically tailored to their needs. The main purpose of this application is to assist job seeking users in developing career focused skills which are identified as deficient and which will not likely be taught to a sufficient level through their current degree program. Additionally, the application will facilitate the tracking of student and alumni learning progress as they develop target competencies.

This application should serve as a streamlined intermediary between the user and learning resources. This means the application should eliminate as many obstacles between the user and their tailored resources as possible. This application should NOT become another program the user has to manage ad nauseam. Instead, the application must prioritize accessibility and present information in manageable, digestible segments to avoid user overwhelm.

This application will NOT be home to custom modules or lessons. Rather, it will help determine which pre-existing online resources would be most relevant to a user. This determination will be extrapolated from skills the student has indicated they would like to improve by a specified date (e.g. date of graduation).

Once the student or alumni has verified their interest in specific learning resources, the application should assist the user in scheduling time to complete tasks by a desired date. Additionally, the application should provide the user with a gamified progress tracking experience (e.g. experience bar, achievements, badges, etc.) to incentivize learning.

## 1.4 References

The following sources should be referenced while building and maintaining The *CaLLP* application to ensure all members of the intended user-base can easily and securely benefit from the services it provides:

- Accessibility Best Practices for Web Development:
  - <https://www.w3.org/TR/2024/REC-WCAG22-20241212/>
  - <https://developer.mozilla.org/en-US/docs/Web/Accessibility>
- Account Linking Best Practices for Cross-platform Authentication:
  - <https://www.loginradius.com/blog/identity/account-linking-social-login-ux>
- Cal Poly Humboldt Single Sign On (SSO) Information:
  - <https://www.humboldt.edu/its/saml-sso-request-form>

## 2. Overall Description

### 2.1 Project Perspective

The *CaLLP* application is a new product intended to enhance the career outlooks of the students and alumni of Cal Poly Humboldt.

### 2.2 Project Functions

This section provides a high-level overview of the application's primary functionality. The complete feature set will be more technically described in the **System Features** section to follow.

- **User Account Management:**
  - Allow students, alumni, and authorized administrators to securely access functionality appropriate to their roles.
- **User Onboarding and Career Profiling:**
  - Gather information about users' degree programs, career interests, existing strengths, desired skills, skill deficiencies, and target completion dates.
- **Learning Goal Development:**
  - Help student and alumni users identify and confirm career-focused skills and competencies they would like to develop.
- **Learning Resource Discovery:**
  - Connect users with existing and relevant learning resources from external educational platforms based on their identified learning goals and skill-development needs.
- **Learning Pathway Planning:**
  - Organize selected skills and learning resources into manageable pathways geared toward users' career and learning goals.
- **Availability-Based Scheduling:**
  - Assist users in scheduling learning activities according to their weekly availability and desired goal-completion dates.
- **Progress Tracking:**
  - Assist users in recording and monitoring their progress as they develop skills and competencies.
- **Reminders and Notifications:**
  - Provide configurable notifications to remind users of planned learning activities, deadlines, and progress milestones.

- **Gamification and Motivation:**
  - Provide progress indicators and other gamification elements, such as achievements or badges, to encourage continued participation.
- **Student and Alumni Dashboard:**
  - Present learning goals, progress, schedules, and relevant resources through a centralized and accessible interface.
- **Career Resource Integration:**
  - Connect users with relevant external career and hiring resources (such as LinkedIn, Handshake, Indeed, etc) where appropriate.
- **Learning Resource Integration:**
  - Connect users with relevant external educational and learning resources (such as Coursera, Udemy, TryHackMe, etc.) where appropriate.
- **Administrative Management:**
  - Provide authorized administrators with tools to manage application content, resources, and other configurable information such as user account deletion and suspension.
- **User Analytics:**
  - Provide authorized administrators with aggregated information about application usage, such as commonly selected skills, careers, learning goals, and searches.
- **Responsive and Accessible Interaction:**
  - Provide application functionality across common desktop, laptop, tablet, and mobile web environments while supporting established web accessibility practices.

## 2.3 User Classes and Characteristics

This section seeks to describe the expected user types who will most likely use and benefit from this application.

- **Student:**
  - A student user is currently enrolled at Cal Poly Humboldt and is seeking a degree or actively pursuing a learning pathway through the university. They are actively working to develop career-focused skills and competencies to augment their degree program. Students may or may not be actively seeking employment.
- **Alumni:**
  - An alumni user is a graduate of Cal Poly Humboldt. They are actively working to develop their career-focused skills and competencies. Alumni may or may not be actively seeking employment.

- **Administrator:**
  - A trusted Cal Poly Humboldt employee responsible for account management and interested in the application's efficacy. The administrator hopes to use available analytics to monitor and enhance student and alumni user experiences.

## 2.4 User Stories

- **Student:**
  - US-1.1 **As a user**, I want to see a selection of skills available for me to learn when I open the homepage **so I can** avoid analysis paralysis.
  - US-1.2 **As a user**, I want to complete the onboarding process **so I can** receive personally tailored skills to train.
  - US-1.3 **As a user**, I want to connect my CaLLP account to my coursera account **so I can** see my progress automatically update.
- **Alumni:**
  - US-2.1 **As an alumnus**, I want to see how far I have progressed **so I can** see where I can further improve.
  - US-2.2 **As an alumnus**, I want to see the progress of current students **so I can** offer appropriate advice.
- **Administrator:**
  - US-3.1 **As an administrator**, I want to monitor content **so I can** ensure it is appropriate, accurate, and useful for student progress.
  - US-3.2 **As an administrator**, I want to edit student progress plans **so I can** tailor plans to student needs.
  - US-3.3 **As an administrator**, I want to track user progress and activity **so I can** understand how effective this tool is for students and alumni.

## 2.5 Operating Environment

The *CaLLP* application will operate as a web-based application accessible via modern web browsers on internet enabled devices such as desktops, laptops, tablets, and mobile devices. Client-side operating environments are expected to include:

- Internet enabled desktop and laptop computers running Windows OS, macOS, or Linux.
- Internet enabled tablets and mobile devices running Android or iPadOS/iOS.
- Web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and/or Apple Safari.

## 2.6 Design and Implementation Constraints

The following is a list of project constraints which are anticipated to limit the design and implementation of the *CaLLP* application:

- **Time Constraints:**
  - The development cycle of the first deliverable prototype for this application is limited to approximately three months.
- **Team Constraints:**
  - The team consists of only four members.
  - The team members are NOT able to develop the project on a full-time basis due to other educational and career commitments.
- **Budget Constraints:**
  - The budget for this application's first prototype is effectively \$0.
- **Software Constraints:**
  - Project development tooling is limited to free services.
  - Software implementation is limited by the team's ability to learn and incorporate said software into the project.

## 2.7 Documentation

The following is a list of documentation resources to be delivered with the first prototype of the application:

- **Software Design Specification:**
  - A detailed technical blueprint which defines how *CaLLP* will be developed from an architectural design perspective.
- **User Manual:**
  - Contains guidance for users of the application.
- **Programmer Manual:**
  - Contains guidance for developers of the application.

## 2.8 Assumptions and Dependencies

The following are the current working assumptions and dependencies on which the *CaLLP Application* functionality relies:

- **Application Program Interface (API) Availability:**
  - It is assumed either third party APIs or APIs developed in-house can support the desired connectivity between the application and career/education platforms for progress-tracking.

- **External Resources Availability:**
  - It is expected external resource availability will be dynamic and the application architecture will have to account for resources which become unavailable, outdated, or are relocated.
- **Authentication Service Availability:**
  - If third-party authentication services are used, the application's authentication functionality will be dependent on the availability of those services.
- **Database Services Availability:**
  - If third-party database services are used, the application's database architecture and functionality will be dependent on the availability of those services.
- **Web Hosting Availability:**
  - The application's functionality will depend on the reliability and availability of the provider used for hosting the web application.

## 3. System Features

### 3.1 User Authentication

- **Description and Priority:**
  - The User Authentication feature allows the user to securely login to their *CaLLP* account.
  - This feature is **High priority** since user specific accounts are required to facilitate a personalized experience. Additionally, authentication is necessary to ensure the privacy and security of the user's data.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user attempts to login from the web-application
    - **Response:** a user is redirected to their Cal Poly Humboldt login for authentication.
  - **Stimulus:** a user submits invalid Cal Poly Humboldt credentials.
    - **Response:** the user is presented with a notification of the invalid credentials and is denied access to the *CaLLP* account.
  - **Stimulus:** a user submits valid Cal Poly Humboldt credentials.
    - **Response:** the user is logged into their *CaLLP* account and redirected to their user applicable dashboard.
  - **Stimulus:** a user selects the logout option from their *CaLLP* account.
    - **Response:** the user's account session is terminated and they are returned to the *CaLLP* home/login web page.
- **Functional Requirements:**
  - **FR-3.1.1:** The system shall require users to be authenticated before providing restricted access to said users.
  - **FR-3.1.2:** The system shall deny access to users who provide invalid authentication credentials.
  - **FR-3.1.3:** The system shall provide restricted access to users who provide valid authentication credentials
  - **FR-3.1.4:** The system shall restrict access to role-specific functionality, determined by the user's role (i.e. student, administrator)
  - **FR-3.1.5:** The system shall allow users to securely terminate their session (log out).

## 3.2 User Career and Skill Profiling

- **Description and Priority:**
  - The User Career and Skill Profiling feature creates a user specific profile which can be utilized to personalize the user's experience while interacting with other *CaLLP* features.
  - This feature is a **High priority** as the primary function of the application relies heavily on profile information being accurate and accessible.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user creates a new account on the *CaLLP* application.
    - **Response:** the user is prompted to begin an onboarding questionnaire.
  - **Stimulus:** a user submits answers to profile specific questions while completing the onboarding questionnaire or when prompted elsewhere in the application.
    - **Response:** the application leverages its database architecture to store this profile information and links it specifically to the applicable user account for later retrieval.
- **Functional Requirements:**
  - **FR-3.2.1:** The system shall retrieve pre-determined questions from the applications database to populate the onboarding questionnaire and other profiling question prompts presented in the application.
  - **FR-3.2.2:** The system shall store the answers to profiling questions.
  - **FR-3.2.3:** The system shall store profile specific information in a well structured database.
  - **FR-3.2.4:** The system shall ensure the appropriate profile information is properly associated with user accounts.

## 3.3 Learning Pathway Management

- **Description and Priority:**
  - The Learning Pathway Management feature allows the user to establish learning objectives in order to build skill and competency development pathways tailored specifically to the user managing them.
  - This feature is **High priority** since the *CaLLP* application's efficacy is directly related to its ability to connect the user with resources which support user defined goals.

- **Stimulus/Response Sequences:**

- **Stimulus:** a user navigates to the Learning Pathway tab or page via the web application interface for the first time.
  - **Response:** a list of suggested skill and competency objectives which align with the current state of the user Career and Skill profile information is displayed (left pane). The list is sorted by perceived relevance to the user. The right pane displays a list of suggested pathways, pre-populated with no more than three learning objectives from the left pane. None of these pathways are active. Also in the right pane, an option to create a custom pathway is displayed.
- **Stimulus:** a user navigates to the Learning Pathway tab or page via the web application interface after already setting up at least one pathway:
  - **Response:** a list of suggested skill and competency objectives which align with the current state of the user career and skill profile information is displayed (left pane). The list is sorted by perceived relevance to the user and does not include objectives already listed in active pathways. The right pane shows active pathways at the top of the list with options to edit them. Additional suggested pathways (no more than three) are displayed at the bottom of the right pane. An option to add a custom pathway is also displayed.
- **Stimulus:** a user selects one or more objectives to add to one the active Learning Pathways.
  - **Response:** The system updates relevant database tables to refine the applications search and suggestion features to match the user selections.
- **Stimulus:** a user deselects/removes one or more objectives from one of the active Learning Pathways.
  - **Response:** The system updates relevant database tables to refine the applications search and suggestion features to match the user's updated pathway.

- **Functional Requirements:**

- **FR-3.3.1:** The system shall allow users to add, delete, and edit pathways from a single web page.
- **FR-3.3.2:** The system shall display top skill and competency objective suggestions which have not yet been selected by the user.
- **FR-3.3.3:** The system shall display currently selected skill and competency objectives within the currently active pathway.

### 3.4 Learning Resource Discovery

- **Description and Priority:**
  - The Learning Resource Discovery Feature gathers and presents relevant learning resources to the user in a manageable and helpful manner.
  - This feature is **High priority** as it is one of the primary functions of the CaLLP application.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user views learning resources associated with a selected skill, competency objective, or Learning Pathway.
    - **Response:** the system retrieves and displays a manageable selection of relevant learning resources, prioritized according to their relevance to the user's selected objectives and available profile information.
  - **Stimulus:** a user selects a displayed learning resource.
    - **Response:** the system provides available resource information and allows the user to access the applicable external learning resource.
- **Functional Requirements:**
  - **FR-3.4.1:** The system shall retrieve learning resources relevant to user-selected skills and competency objectives.
  - **FR-3.4.2:** The system shall prioritize retrieved learning resources according to their relevance to the user.
  - **FR-3.4.3:** The system shall display resource information including the resource name, source, and applicable external link.
  - **FR-3.4.4:** The system shall allow users to access external learning resources from the CaLLP application.
  - **FR-3.4.5:** The system shall limit initially displayed resource suggestions to a manageable selection.

### 3.5 Availability, Scheduling, and Calendar Management

- **Description and Priority:**
  - The Availability, Scheduling and Calendar Management feature allows the user to provide times and dates they are available to pursue skill and competency development. Using the user provided availability and pathway information, this feature suggests viable scheduling options to meet the user's defined target completion date.

- This feature is **Medium priority** as it is secondary to providing the user with resources but essential to guiding the user in accomplishing objectives by a specific date.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user provides or updates their weekly availability and target completion dates.
    - **Response:** the system stores the provided scheduling information and uses it to determine available periods for learning activities.
  - **Stimulus:** a user requests scheduling assistance for an active Learning Pathway.
    - **Response:** the system presents a proposed schedule which distributes applicable learning activities across the user's available time and target completion period.
  - **Stimulus:** a user accepts or modifies a proposed schedule.
    - **Response:** the system updates the user's calendar to reflect the accepted learning schedule.
- **Functional Requirements:**
  - **FR-3.5.1:** The system shall allow users to define and modify their weekly availability.
  - **FR-3.5.2:** The system shall allow users to define target completion dates for applicable Learning Pathways or objectives.
  - **FR-3.5.3:** The system shall generate scheduling suggestions based on user availability and target completion dates.
  - **FR-3.5.4:** The system shall allow users to accept or modify suggested learning schedules.
  - **FR-3.5.5:** The system shall display scheduled learning activities through a user calendar.

## 3.6 Student and Alumni Dashboard

- **Description and Priority:**
  - The Student and Alumni Dashboard feature centralizes and presents the *CaLLP* application's functionality in an appealing and accessible manner.
  - This feature is **High priority** as it abstracts the application's primary functions in one centralized location to maximize user experience and engagement.

- **Stimulus/Response Sequences:**
  - **Stimulus:** an authenticated student or alumni user navigates to their dashboard.
    - **Response:** the system displays a centralized overview of information relevant to the user, including active Learning Pathways, progress, upcoming scheduled activities, applicable learning resources, linked accounts, and leader/user feeds (if enabled in settings).
  - **Stimulus:** a user selects an available dashboard component or action.
    - **Response:** the system displays the corresponding information or directs the user to the applicable CaLLP feature.
- **Functional Requirements:**
  - **FR-3.6.1:** The system shall provide authenticated student and alumni users with a personalized dashboard.
  - **FR-3.6.2:** The system shall display active Learning Pathways and associated objectives on the user dashboard.
  - **FR-3.6.3:** The system shall display current learning progress on the user dashboard.
  - **FR-3.6.4:** The system shall display upcoming scheduled learning activities on the user dashboard.
  - **FR-3.6.5:** The system shall provide access to relevant learning resources and primary user functionality from the dashboard.

### 3.7 Progress Tracking and Gamification

- **Description and Priority:**
  - The Progress Tracking and Gamification feature provides the user with reward style incentives to begin and continue developing skills and competencies.
  - This feature is **Medium priority** as it is not essential for the *CaLLP* application to provide its services. However, this feature contributes significantly to increasing user engagement and satisfaction which may ultimately translate to users accomplishing their goals.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user records progress toward a Learning Pathway objective or progress is received through an integrated educational platform.
    - **Response:** the system updates the user's applicable progress information and reflects the change through associated progress indicators.

- **Stimulus:** updated user progress satisfies the requirements for a defined achievement or milestone.
  - **Response:** the system awards the applicable achievement, badge, or other gamification indicator and presents it to the user.
- **Functional Requirements:**
  - **FR-3.7.1:** The system shall store user progress associated with Learning Pathways and objectives.
  - **FR-3.7.2:** The system shall allow users to manually update applicable learning progress.
  - **FR-3.7.3:** The system shall visually display progress toward active learning objectives and pathways.
  - **FR-3.7.4:** The system shall identify when user progress satisfies defined milestones or achievements.
  - **FR-3.7.5:** The system shall display earned achievements, badges, or other implemented gamification indicators.

### 3.8 Notifications

- **Description and Priority:**
  - The Notifications feature provides the user with reminders to keep them on track to accomplish pathway objectives by target milestone and completion dates.
  - This feature is **Low priority** since it is not critical to the primary functionality of the *CaLLP* application. It is however an assistive feature which will likely improve user experiences and increase engagement.
- **Stimulus/Response Sequences:**
  - **Stimulus:** a user configures their notification preferences for scheduled learning activities, milestones, or completion dates.
    - **Response:** the system stores the user's notification preferences and applies them to applicable future notifications.
  - **Stimulus:** a scheduled activity, milestone, or completion date meets the conditions established by the user's notification preferences.
    - **Response:** the system provides the applicable reminder through a supported notification method.
- **Functional Requirements:**
  - **FR-3.8.1:** The system shall allow users to enable or disable configurable notifications.
  - **FR-3.8.2:** The system shall store user-specific notification preferences.

- **FR-3.8.3:** The system shall generate reminders for applicable scheduled learning activities, milestones, and target completion dates.
- **FR-3.8.4:** The system shall only provide configurable notifications according to the user's stored preferences.

### 3.9 Employment and Educational Platform Integration

- **Description and Priority:**

- The Employment and Educational Platform Integration feature allows the user to link their active accounts on other platforms (e.g. LinkedIn, Coursera, etc.) for quick logins. This feature will streamline transitions to learning resources provided by these platforms.
- This feature is **Medium priority** as it's not critical for directing users to pertinent resources. This feature removes obstacles and extra steps between the user and the learning resources.

- **Stimulus/Response Sequences:**

- **Stimulus:** a user chooses to link their CaLLP account with a supported employment or educational platform.
  - **Response:** the system directs the user through the applicable external authorization process and, upon successful authorization, associates the external account with the user's CaLLP account.
- **Stimulus:** a user accesses functionality associated with a successfully linked platform.
  - **Response:** the system uses the authorized integration to provide supported platform functionality or information without requiring the user to manually establish a new connection.

- **Functional Requirements:**

- **FR-3.9.1:** The system shall allow users to initiate account linking with supported external platforms.
- **FR-3.9.2:** The system shall require successful external authorization before linking an external account.
- **FR-3.9.3:** The system shall securely associate authorized external platform accounts with the applicable CaLLP user.
- **FR-3.9.4:** The system shall provide supported integration functionality for successfully linked platforms.
- **FR-3.9.5:** The system shall allow users to disconnect previously linked external platform accounts.

### 3.10 Administrative Actions

- **Description and Priority:**

- The Administrative Actions feature provides administrators with the ability to create, delete, suspend, and reactivate user accounts. This feature also allows administrators to view most user account information for the intended purpose of assisting users with any account issues. Additionally, this feature allows administrators to manage the *CaLLP* application's content and resources.
- This feature is **High priority** since it is crucial to have the ability to manage user accounts to ensure the best possible user experience for anyone who chooses to engage with the *CaLLP* application.

- **Stimulus/Response Sequences:**

- **Stimulus:** an authorized administrator selects a user account for administrative management.
  - **Response:** the system displays the account information and administrative actions available to the administrator.
- **Stimulus:** an authorized administrator creates, modifies, suspends, reactivates, or deletes applicable application content, resources, or user accounts.
  - **Response:** the system validates the requested action, updates the applicable stored information, and provides the administrator with confirmation of the result.

- **Functional Requirements:**

- **FR-3.10.1:** The system shall restrict administrative actions to authorized administrators.
- **FR-3.10.2:** The system shall allow authorized administrators to view user account information necessary for account management.
- **FR-3.10.3:** The system shall allow authorized administrators to create, suspend, reactivate, and delete applicable user accounts.
- **FR-3.10.4:** The system shall allow authorized administrators to add, edit, and remove application-managed content and learning resources.
- **FR-3.10.5:** The system shall provide confirmation when an administrative action succeeds or fails.

### 3.11 Administrative Dashboard

- **Description and Priority:**
  - The Administrative Dashboard feature centralizes and presents the *CaLLP* application's administrative functions in an efficient and accessible manner.
  - This feature is **High priority** as it abstracts crucial administrative functionalities into one centralized location for ease of use.
- **Stimulus/Response Sequences:**
  - **Stimulus:** an authorized administrator navigates to the Administrative Dashboard.
    - **Response:** the system displays administrative tools, application information, and available management functionality in a centralized interface.
  - **Stimulus:** an administrator selects an available dashboard component or administrative function.
    - **Response:** the system displays the applicable information or management interface while maintaining administrator-specific access restrictions.
- **Functional Requirements:**
  - **FR-3.11.1:** The system shall provide authorized administrators with an Administrative Dashboard.
  - **FR-3.11.2:** The system shall provide access to user account management from the Administrative Dashboard.
  - **FR-3.11.3:** The system shall provide access to application content and resource management from the Administrative Dashboard.
  - **FR-3.11.4:** The system shall provide access to available User Analytics from the Administrative Dashboard.
  - **FR-3.11.5:** The system shall restrict Administrative Dashboard functionality from non-administrator users.

### 3.12 User Analytics

- **Description and Priority:**
  - The User Analytics feature provides authorized administrators with aggregated information about application usage, such as commonly selected skills, careers, learning goals, and searches.
  - This feature is **Low priority** since it is not required for application functionality.

- **Stimulus/Response Sequences:**
  - **Stimulus:** an authorized administrator accesses User Analytics from the Administrative Dashboard.
    - **Response:** the system retrieves and presents aggregated application usage information without requiring the administrator to inspect individual user records.
  - **Stimulus:** an administrator selects an available analytics category or filtering option.
    - **Response:** the system updates the displayed analytics to represent the selected category or available filtering criteria.
- **Functional Requirements:**
  - **FR-3.12.1:** The system shall collect application usage information necessary to support defined administrative analytics.
  - **FR-3.12.2:** The system shall aggregate collected usage information for administrative presentation.
  - **FR-3.12.3:** The system shall provide analytics for commonly selected skills, careers, learning goals, and searches.
  - **FR-3.12.4:** The system shall allow administrators to view available analytics through the Administrative Dashboard.
  - **FR-3.12.5:** The system shall restrict access to User Analytics to authorized administrators.

### 3.13 Non-User Specific Landing Page

- **Description and Priority:**
  - The Non-User Specific Landing Page feature provides prospective and unauthenticated users with immediate access to a manageable selection of learning and career resources without requiring a user-specific profile. The landing page provides responsive access to default resources and allows users to browse resources according to categories maintained by the application.
  - This feature is **High priority** since it will be the first step in integration and provide the longest lasting example of our provided resources to all prospective users

- **Stimulus/Response Sequences:**

- **Stimulus:** a prospective or unauthenticated user navigates to the CaLLP landing page.
  - **Response:** the system displays a responsive landing page containing a manageable selection of default resources, available resource categories, and access to user authentication.
- **Stimulus:** a user selects a category from the available categories and initiates a search.
  - **Response:** the system retrieves and displays resources and career information relevant to the selected category.
- **Stimulus:** a user selects one of the resources displayed on the landing page.
  - **Response:** the system provides access to the applicable external resource without requiring the user to create or authenticate a CaLLP account.

- **Functional Requirements:**

- **FR-3.13.1:** The system shall display a selection of default learning, career, and lifelong-development resources on the landing page.
- **FR-3.13.2:** The system shall retrieve available resource categories from application-managed data.
- **FR-3.13.3:** The system shall allow users to select a resource category and request category-specific results.
- **FR-3.13.4:** The system shall display resources and career information relevant to the selected category.
- **FR-3.13.5:** The system shall allow users to access publicly available landing-page resources without authentication.

## 4. Nonfunctional Requirements

### 4.1 Security Requirements

The following is a list of security considerations and requirements in order to protect the privacy and security of users of the *CaLLP* application:

- Use *OAuth 2.0* and Proof Key for Code Exchange (*PKCE*) protocols to enable cross-platform authorization without sharing or storing passwords.
- Verify the JSON Web Token (*JWT*) signature, issuer, and audience to prevent replay attacks and token manipulation.
- Only link accounts in which the platform provider has designated the user's email as verified.
- Require 2FA (two factor authentication) before linking new platform accounts.
- Do NOT automate account linking. This should be user initiated only.
- Consider logging account linking activity with applicable metadata.

### 4.2 Software Quality Attributes

The following is a breakdown of the *CaLLP* application's quality attributes with a focus on effective system operation:

- **Usability:**
  - The application shall present information and navigation to the user in a consistent, intuitive, and accessible manner.
  - The application shall present user specific information, such as resources, objectives, and progress, in digestible manner.
  - Appropriate feedback shall be provided to users to indicate if actions within the application succeeded or failed.
- **Accessibility:**
  - The application shall conform to the Web Content Accessibility Guidelines (*WCAG*) 2.2 Level AA accessibility standard.
  - User-interface controls shall provide accessible labeling for assistive technologies.
- **Maintainability:**
  - Source code shall be accompanied by documentation and or include in-line comments necessary for future developers to understand non-obvious functionality.

- **Reliability:**
  - Invalid user input shall not cause the application to enter an unrecoverable state.
  - The application shall preserve previously committed user information when a subsequent operation fails.
  - Failure of an external learning or employment service shall not prevent users from accessing application functionality that does not depend upon that service.
- **Interoperability:**
  - The application shall support integration with external service platforms through secure interfaces.
  - Documentation will be provided to detail how integrations were implemented and provide guidance on how they should be maintained.
- **Portability:**
  - The application shall provide its primary functionality through supported desktop and mobile web browsers.
  - The application shall not require installation of platform-specific client software.
  - The user interface shall adapt to desktop, tablet, and mobile viewport sizes without loss of primary functionality or accessibility.
- **Testability:**
  - Functional components shall provide defined inputs and observable outputs to support manual and automated testing approaches.
  - Automated tests shall be executable without requiring modification of production source code.

## 5. Software Development Process

### 5.1 Project Development Model

Our team has selected the **Incremental Model**, with emphasis on prototyping, for our development process. We currently favor this model for CaLLP development because it allows us to develop the application incrementally over the next three months. This incremental approach gives us the flexibility to build toward our client's vision in stages of increasing complexity. As each stage is completed, we can show prototype progress to the client and incorporate feedback as we develop the application into its final form.

### 5.2 Functional Area to be Developed First

The first functional area our team will be working on first is the **Non-user Specific Landing Page** in order to prototype the client's primary feature and functionality sets. This will also allow the team to fine tune accessibility, design, and navigation which is aligned with the client's vision. Once this functional area is developed we will be able to expand the feature sets more easily to meet the client's full feature requirements.

## **Appendix A: Team Member Roles and Responsibilities**

### **Engineering Lead and Systems Analyst: Roselie Freixas**

- Lead overall technical planning and coordinate development activities across the team.
- Serve as the primary liaison between the development team and project sponsor/client.
- Maintain and refine the Software Requirements Specification (SRS), Software Design Specification (SDS), User Manual, and Programmer Manual.
- Coordinate system-level architecture and ensure frontend, backend, and QA designs remain aligned
- Monitor project scope and identify requirements changes or potential scope creep.
- Facilitate technical decision-making and resolution of cross-functional issues.
- Coordinate task prioritization, development milestones, and integration activities.
- Maintain and administrate the Github Organization, Repository, and Project.
- Support code/design reviews and contribute to general application development and testing as needed.

### **Frontend Architect: Skyler Hunsinger**

- Lead the design and architecture of the application's client-side components.
- Establish frontend structure, component organization, and development conventions.
- Lead implementation of responsive user interfaces and navigation.
- Ensure frontend designs meet established usability and accessibility requirements.
- Promote consistency in interface behavior and presentation throughout the application.
- Document frontend architecture and relevant interface designs.
- Maintain and refine the User Manual documentation and Programmer Manual.
- Coordinate frontend integration with backend services and APIs.
- Contribute to general application development and testing as needed.

**Backend Architect: Clay Cordary**

- Lead the design and architecture of server-side application components.
- Design application data models, persistence mechanisms, and database interactions.
- Lead implementation of authentication, authorization, and other server-side functionality.
- Design integration mechanisms for required third-party services and APIs.
- Ensure backend components support established security, privacy, reliability, and performance requirements.
- Document backend architecture, data designs, and software interfaces.
- Maintain and refine the Programmer Manual documentation.
- Establish backend service and API structures used by the frontend.
- Contribute to general application development and testing as needed.

**Quality Assurance Architect: Jerry Huang**

- Lead development and maintenance of the project's overall testing and quality-assurance strategy.
- Develop test plans and test cases from functional and nonfunctional requirements.
- Evaluate implemented functionality against established requirements and acceptance criteria.
- Coordinate automated testing practices and supporting test infrastructure.
- Identify, document, track, and verify resolution of software defects.
- Coordinate regression testing following changes to existing functionality.
- Document testing procedures, results, known issues, and quality-related findings.
- Maintain and refine the Programmer Manual documentation.
- Contribute to general application development and testing as needed.

## **Appendix B: Glossary**

API – Application Program Interface

CaLLP – Career & Lifelong Learning Pathway

JWT – JSON Web Token

PKCE – Proof Key for Code Exchange

SDS - Software Design Specification

SRS - Software Requirements Specification

WGAC – Web Content Accessibility Guidelines

## Appendix C: AI Assistance Disclosure

**AI Assistance Disclosure:** OpenAI's ChatGPT was used as a supplementary tool during preparation of this document to assist with brainstorming, terminology clarification, document organization, and drafting or refining selected language. The authors retain responsibility for requirements analysis, project-specific decisions, verification of content, and the final form of the document.